# Preliminary Project - AIS2901

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# 1 Introduction

The purpose of this bachelor project is to contribute to the further development of Sub Horizon as a technical student organization with a focus on maritime robotics. Previously, Sub Horizon has its focus on ROVs, but this year the organization has expanded its scope by building a test boat that will function as a development platform for autonomous vessels on water. The task will lay the foundation for hardware, software and autonomous operation of boats for Sub Horizon. The bachelor's group will also have the opportunity to test and demonstrate the solutions in practice through participation in Autodrone 2026, which will provide valuable experience for the organization.

# 2 Concepts

- USV - Unmanned Surface Vehicle
- ASV - Autonomous Surface Vehicle
- HSE - Health, Safety, and Environment

# 3 Project organization

## 3.1 Project group

| Candidate | Student numbers |
|---|---|
| Liam Harris Følsvik | 107210 |
| Ørjan Tennøy Lynnes | 599304 |

## 3.2 Responsibilities

### 3.2.1 Working Hours

Both members of the group have agreed to a working time of 8 hours per day, working on weekdays. Days working on INGA2301 are to be agreed too between the members.

### 3.2.2 Meeting / Secretary

Liam has responsibility for documenting the topics discussed during each meeting. Any formal meetings like supervisor meetings are to be made in Teams so that everyone attending gets an invite.

### 3.2.3 Submissions

Ørjan has responsibility for handling the deliverables, making sure they are correctly formatted and that they are delivered on time.

Liam has responsibility for ensuring that the status report work is initiated and completed.

### 3.2.4 Notification in case of absence

Both members have the responsibility to notify the other team member in case of unexpected absences.

### 3.2.5 Document handling

- Overleaf is used for the status reports and for the Bachelors thesis report.
- GitHub is used for version control of code. The code should generally include some form of documentation. The code may be across several repositories featuring one main repository that collects all of the repositories.
- The worklog, HSE and Gantt is documented in Excel. The document should be updated each work day and should document the activities worked on, hours spent and a update of the Gantt diagram for the corresponding activity.
- Visio may be used for developing diagrams, such as control schemes, system overviews and more.

### 3.2.6 Communication

- Teams and outlook is used for communication with the supervisors.

- Discord is used for sharing resources and general communication between the group members.

# 4 Agreements

## 4.1 Workplace

- The Sub-Horizon office is used as a office and as a workshop when working on the project.
- The location of practical tests of the USV/ASV may be done at different places such as Gardsvika or outside of the museum (Nørvasundet).

## 4.2 Resources

- Components and various resources are covered by Sub-Horizon, as the boat is Sub-Horizons property and is to be used for various testing after the bachelor project is completed.

## 4.3 Group norms - collaboration rules - attitudes

The group should be respective of one and another and maintain good attitudes. The members should attempt to motivate and push the other member such that the pseudo deadlines are met.

An important value for the engineers is thoroughly documented code with user manuals. This makes it easier for others to join in on the project and be able to quickly understand the structure and the principles in the project.

Honesty and transparency about issues encountered during the project should also handled appropriately. This involves being honest about the identified weaknesses and the quality of the work performed. This should generally be discussed among the projects members and be documented. This is important to be able to produce high quality solutions that can be revisited and improved in the future.

# 5 Project Description

## 5.1 Problem formulation

Sub Horizon is set to compete in the Autodrone 2026 competition. The competition gives Sub Horizon a bigger goal to work towards. The competition involves completing several challenges with a ASV as mentioned below:

- Speed mission - The seadrone is timed as it speeds through a set of start buoys, then navigates around the turning buoy, and returns through the start buoys again[1].
- Collision avoidance - The seadrone must pass through a trafficked area avoiding another vessel according to navigational collision avoidance regulations[1].
- Obstacle Channel - the Obstacle Channel mission proves the seadrone's capability to sense and maneuver through a new path while avoiding obstacles in the way[1].

- Docking - The seadrone must try to dock to an area at a quay marked by two green buoys[1]. Figure 1 displays the general layout of the buoys in each mission.

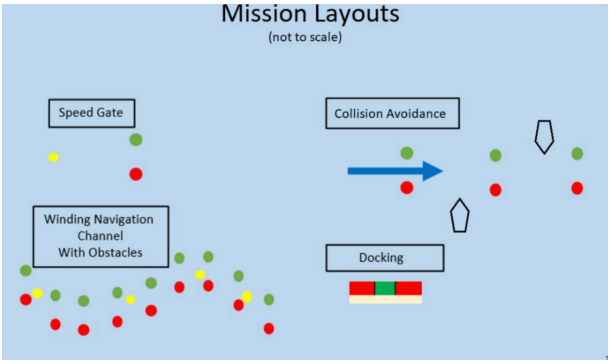



Figure 1: Autodrone layout of challenges

The USV/ASV is developed for the student organization Sub Horizon to gain further insights into maritime robotics. The bachelor project aims to further improve Sub Horizon as a student organization by developing a baseline that is capable of both dynamic positioning and performing autonomous tasks using visual data. Currently Sub Horizon lacks software capable of solving the challenges previously described. Consequently, the problem is to design and implement the essential hardware and software framework to allow for autonomous control.

The resultant goal of the project is to deliver a ASV capable of competing in the Autodrone 2026 competition, and for that to be possible then the following systems must be in place:

- Operational USV - electronics & communication.
- Simulation of the USV.
- GUI capable of selecting different modes, displaying map with waypoints along with other useful data and controls.
- Control system that is capable of ensuring that the USV/ASV follows or maintains a coordinate by using (DP).
- Capable vision system that is able to detect and locate buoys of three different colors; red, green and yellow. Additionally the vision system should be able to detect whenever another boat is on a collisions course.
- Path planning system able to create a path that avoids collisions.

In order to work towards the projects goals the Gantt diagram should be followed with hardly any deviation.

## 5.2 Requirements for solution or project result – specification

The first and foremost goal is to get the USV to a operational level, this involves building the hull, connecting the electronics and setting up communication between the boat and a ground control station.

The dynamic positioning system is extremely important for the success of the project, without it the USV won't be able to follow any set of trajectories/coordinates given to it. The vision system is the key to gathering information in order to solve all of the tasks. Every challenge requires the vision system to be able to detect a set of buoys, and pinpoint the objects location. Additionally the vision system should be able to detect a incoming boat such that the USV can give way to it.

Additional requirements involve following the given rules in the AutoDrone competition:

- **Autonomy:** Drone shall be fully autonomous and shall have all autonomy decisions made onboard the ASD.
- **Communication:** The drone cannot send or receive any control information to and from Operators Control Station while in autonomous mode.
- **Deployable:** The ASD should be manually deployable.
- **Energy source:** The drone must be battery powered. All batteries must be sealed to reduce the hazard from acid or caustic electrolytes. The open circuit voltage of any battery (or battery system) may not exceed 60Vdc.
- **Kill Switch:** The drone must have at least one red button located on the drone that, when actuated, must instantaneously disconnect power from all motors and actuators.
- **Wireless Kill Switch:** In an emergency situation the operator control station must be able to actuate the kill switch on board the ASD.
- **Propulsion:** Any propulsion system may be used (thruster, paddle, etc.). However, all moving parts must have protection. For instance, a propeller must be shrouded.
- **Remote-controllable:** The drone must be remote-controllable from an operator control station.
- **Safety:** All sharp, pointy, moving or sensitive parts must be covered and marked.
- **Towable:** The drone must be towable.
- **Visual Feedback:** Visual indication of operating mode. Red indicates that the vehicle is disarmed, Yellow indicates manual operation and green/blue indicates autonomous operation.
- **Weight:** The weight must not weight more than 70 kg.
- **Payload:** The drone must have a place to mount a go-pro action camera with an unobstructed view from the front of the drone.

## 5.3 Planned approaches for the development work – methods

**Physical design and construction** The test boat began development in mid-2025 as a platform for testing future USV development in the Sub-Horizon. It includes four azimuth thrusters, one at each corner, allowing different thruster configurations to be tested.

The boat is a catamaran, chosen for improved stability at this scale and to allow space between the thrusters by placing them at the corners of the hulls. It is powered by large

battery packs designed for long-range or long-time operation, with a total capacity of 1152 Wh.

The electronics are mounted in three separate locations on the boat, the left hull, the right hull, and a small central box. The left hull contains the power delivery components, while the right hull houses the compute hardware, such as the NVIDIA Jetson Orin NX. The central box holds sensors and related components, including the IMU, GPS, stereocamera (Zed 2i), radio receiver, and other sensors. All three boxes are connected using waterproof connectors and are designed to be modular. This allows the system to be easily serviced, or separated into smaller parts for transport.

The GPS antennas are mounted diagonally on the decks of the boat to maximize the distance between them, improving heading accuracy. Two emergency stop buttons are installed at the remaining diagonal deck positions, providing one emergency stop on each side of the boat for easy access. Activating an emergency stop immediately cuts all power to the thrusters and the azimuth servo motors.

**Simulation** The simulation is going to be developed in Stonefish [2]. Stonefish is a simulator created in C++ and allows for simple integration of several sensors and actuators with the ROS2 framework. The simulation should be able to account for the servos, thrusters, stereocamera, GPS and IMU. Furthermore the simulation vehicle should be controlled by ArduPilot in a way that allows it to follow waypoints.

Furthermore the different test scenarios should be added to the simulation. This is achieved by creating individual models of the buoys and adding them to a coordinate such that they are similar to the details given in the task descriptions.

**Dynamic positioning and control** Ardupilot already comes with a control system capable of following waypoints. The main complications arises from the more complex actuator system implemented on the USV (4 Azimuth thrusters). The control strategy and the implementation of it needs to be further tested and experimented with, but there are currently several possible approaches. The first and most recommended approach is to create a LUA script for receiving the steering and throttle commands capable of controlling the azimuth thrusters in a good way. The alternative approach is to lock the azimuth thrusters in a preconfigured position such as OmniX (thrusters oriented at 45 degree angle), this would potentially make the implementation much more simple requiring no modification to the ArduPilot Software. The last approaches involves modifying the Rover configuration or maybe even the Copter configuration (Much more documented), but this will most likely not attempted.

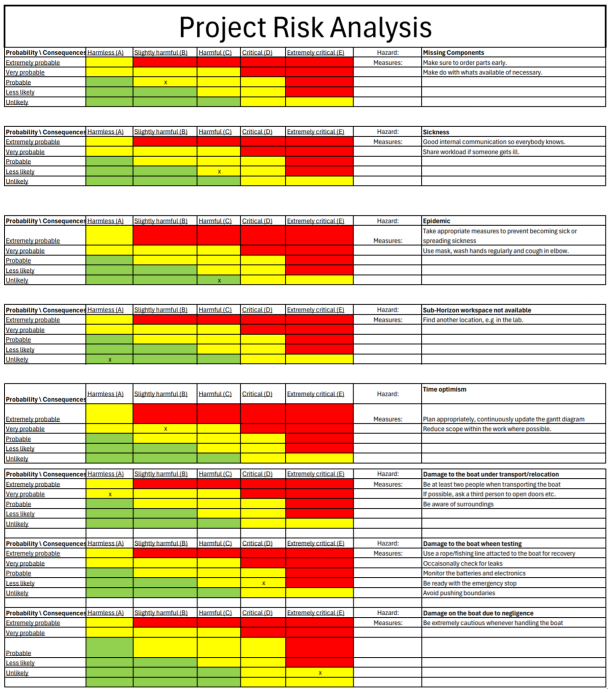
After the Rover is able to follow waypoints it will be neccessary to be able to add waypoints and set waypoints for the boat to follow automatically. In this step it is important to do a lot of testing in ensuring that the USV is able to follow the given coordinates correctly with minimal trajectory error. In order to send the waypoints to the ArduPilot autopilot it is required to generate mavlink commands.

**Vision** The camera available in this project is able to detect the location of the buoys, to achieve this we plan to use the Zed SDK along with their ROS2 wrapper. To properly train the segmentation model, for the buoy detection the plan create a dataset and use annotation tools to mark the buoys. Additionally it is required to test the accuracy of the

depth perception provided by the camera. In order to achieve this we will setup a simple test rig and compare known- and measured distances between a given object.

**Communication** In order to communicate between the vessel and the base station of land a Wi-Fi should be created. This allows for access over SSH to the onboard computer.

## 5.4 Assessment – analysis of risk



Project Risk Analysis

| Probability \ Consequences | Harmless (A) | Slightly harmful (B) | Harmful (C) | Critical (D) | Extremely critical (E) | Hazard: | Missing Components |
|---|---|---|---|---|---|---|---|
| Extremely probable | | | | | | Measures: | Make sure to order parts early. |
| Very probable | | | | | | | Make do with whats available of necessary. |
| Probable | | x | | | | | |
| Less likely | | | | | | | |
| Unlikely | | | | | | | |

| Probability \ Consequences | Harmless (A) | Slightly harmful (B) | Harmful (C) | Critical (D) | Extremely critical (E) | Hazard: | Sickness |
|---|---|---|---|---|---|---|---|
| Extremely probable | | | | | | Measures: | Good internal communication so everybody knows. |
| Very probable | | | | | | | Share workload if someone gets ill. |
| Probable | | | | | | | |
| Less likely | | | x | | | | |
| Unlikely | | | | | | | |

| Probability \ Consequences | Harmless (A) | Slightly harmful (B) | Harmful (C) | Critical (D) | Extremely critical (E) | Hazard: | Epidemic |
|---|---|---|---|---|---|---|---|
| Extremely probable | | | | | | Measures: | Take appropriate measures to prevent becoming sick or spreading sickness |
| Very probable | | | | | | | Use mask, wash hands regularly and cough in elbow. |
| Probable | | | | | | | |
| Less likely | | | | | | | |
| Unlikely | | | x | | | | |

| Probability \ Consequences | Harmless (A) | Slightly harmful (B) | Harmful (C) | Critical (D) | Extremely critical (E) | Hazard: | Sub-Horizon worksgace not available |
|---|---|---|---|---|---|---|---|
| Extremely probable | | | | | | Measures: | Find another location, e.g. in the lab. |
| Very probable | | | | | | | |
| Probable | | | | | | | |
| Less likely | | | | | | | |
| Unlikely | x | | | | | | |

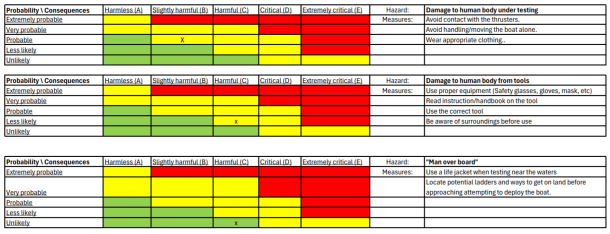
| Probability \ Consequences | Harmless (A) | Slightly harmful (B) | Harmful (C) | Critical (D) | Extremely critical (E) | Hazard: | Time optimism |
|---|---|---|---|---|---|---|---|
| Extremely probable | | | | | | Measures: | Plan appropriately, continuously update the gantt diagram |
| Very probable | | x | | | | | Reduce scope within the work where possible. |
| Probable | | | | | | | |
| Less likely | | | | | | | |
| Unlikely | | | | | | | |

| Probability \ Consequences | Harmless (A) | Slightly harmful (B) | Harmful (C) | Critical (D) | Extremely critical (E) | Hazard: | Damage to the boat under transport/relocation |
|---|---|---|---|---|---|---|---|
| Extremely probable | | | | | | Measures: | Be at least two people when transporting the boat |
| Very probable | x | | | | | | If possible, ask a third person to open doors etc. |
| Probable | | | | | | | Be aware of surroundings |
| Less likely | | | | | | | |
| Unlikely | | | | | | | |

| Probability \ Consequences | Harmless (A) | Slightly harmful (B) | Harmful (C) | Critical (D) | Extremely critical (E) | Hazard: | Damage to the boat wheen testing |
|---|---|---|---|---|---|---|---|
| Extremely probable | | | | | | Measures: | Use a rope/fishing line attacted to the boat for recovery |
| Very probable | | | | | | | Occaisonally check for leaks |
| Probable | | | | | | | Monitor the batteries and electronics |
| Less likely | | | | x | | | Be ready with the emergency stop |
| Unlikely | | | | | | | Avoid pushing boundaries |

| Probability \ Consequences | Harmless (A) | Slightly harmful (B) | Harmful (C) | Critical (D) | Extremely critical (E) | Hazard: | Damage on the boat due to negligence |
|---|---|---|---|---|---|---|---|
| Extremely probable | | | | | | Measures: | Be extremely cautious whenever handling the boat |
| Very probable | | | | | | | |
| Probable | | | | | | | |
| Less likely | | | | | | | |
| Unlikely | | | | | x | | |

Figure 2: Project Risk Analysis



| Probability \ Consequences | Harmless (A) | Slightly harmful (B) | Harmful (C) | Critical (D) | Extremely critical (E) | Hazard: | Damage to human body under testing |
|---|---|---|---|---|---|---|---|
| Extremely probable | | | | | | Measures: | Avoid contact with the thrusters. |
| Very probable | | | | | | | Avoid handling/moving the boat alone. |
| Probable | | X | | | | | Wear appropriate clothing. |
| Less likely | | | | | | | |
| Unlikely | | | | | | | |

| Probability \ Consequences | Harmless (A) | Slightly harmful (B) | Harmful (C) | Critical (D) | Extremely critical (E) | Hazard: | Damage to human body from tools |
|---|---|---|---|---|---|---|---|
| Extremely probable | | | | | | Measures: | Use proper equipment (Safety glasses, gloves, mask, etc) |
| Very probable | | | | | | | Read instruction/handbook on the tool |
| Probable | | | | | | | Use the correct tool |
| Less likely | | | x | | | | Be aware of surroundings before use |
| Unlikely | | | | | | | |

| Probability \ Consequences | Harmless (A) | Slightly harmful (B) | Harmful (C) | Critical (D) | Extremely critical (E) | Hazard: | "Man over board" |
|---|---|---|---|---|---|---|---|
| Extremely probable | | | | | | Measures: | Use a life jacket when testing near the waters |
| Very probable | | | | | | | Locate potential ladders and ways to get on land before approaching attempting to deploy the boat. |
| Probable | | | | | | | |
| Less likely | | | | | | | |
| Unlikely | | | x | | | | |

Figure 3: HSE analysis

## 5.5 Main activities in further work

# Autodrone - Bachelor

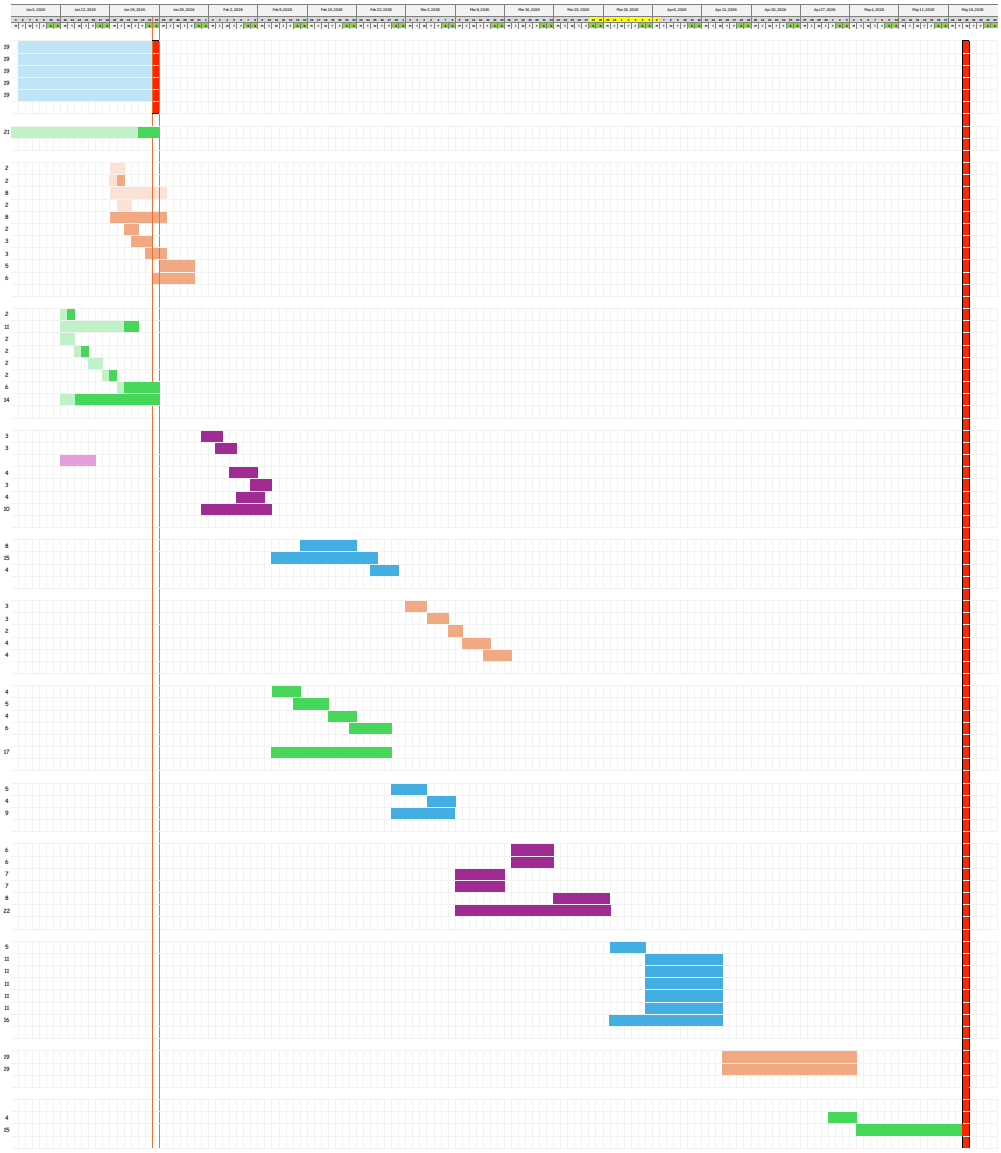
Gruppe 1

SIMPLE GANTT CHARTby Vertex42.com
https://www.vertex42.com/ExcelTemplates/simple-gantt-chart.html

Project start: Tue, 1/6/2026

Display week: 1

| TASK | ASSIGNED TO | PROGRESS | START | END | |
|---|---|---|---|---|---|
| **Preliminary project** | | | | | |
| Create Work Log | Øjan & Liam | 100% | 1/6/26 | 1/24/26 | 19 |
| Create Gantt Plan | Øjan & Liam | 100% | 1/6/26 | 1/24/26 | 19 |
| Perform Project Risk assessment | Øjan & Liam | 100% | 1/6/26 | 1/24/26 | 19 |
| Perform Health Risk assessment (HSE) | Øjan & Liam | 100% | 1/6/26 | 1/24/26 | 19 |
| Write preliminary project report | Øjan & Liam | 100% | 1/6/26 | 1/24/26 | 19 |
| **Final Mechanical Assembly of boat** | | | | | |
| Final Mechanical Assembly of boat | Øjan | 90% | 1/6/26 | 1/25/26 | 21 |
| **Simulator** | | | | | |
| Setup Stonefish project | Øjan | 100% | 1/19/26 | 1/20/26 | 2 |
| Implement the USV/ASV model3 into the simulator environment | Liam | 90% | 1/19/26 | 1/28/26 | 2 |
| Create ROS2 publishers for the simulated sensors (GPS, StereoCamera, IMU, and more) | Liam | 100% | 1/19/26 | 1/26/26 | 8 |
| Create ROS2 subscribers that translates the simulator topics into same topics as the real USV/ASV | Liam | 100% | 1/20/26 | 1/21/26 | 2 |
| Add red, green and yellow buoys to simulation | Øjan | 0% | 1/19/26 | 1/26/26 | 8 |
| Create speed mission | Øjan | 0% | 1/21/26 | 1/22/26 | 2 |
| Create obstacle channel | Øjan | 0% | 1/22/26 | 1/24/26 | 3 |
| Create docking mission | Øjan | 0% | 1/24/26 | 1/26/26 | 3 |
| Create collision avoidance mission | Øjan & Liam | 0% | 1/26/26 | 1/30/26 | 5 |
| Documentation | Liam or Øjan | 0% | 1/25/26 | 1/30/26 | 6 |
| **Electronics** | | | | | |
| Research communication protocol between microcontrollers | Liam | 50% | 1/1/26 | 1/1/26 | 2 |
| Test all electronic components / sensors / actuators | Øjan & Liam | 90% | 1/1/26 | 1/22/26 | 11 |
| Design and build battery box with thruster electronics | Øjan | 100% | 1/1/26 | 1/1/26 | 2 |
| Design emergency stop system | Øjan | 50% | 1/14/26 | 1/15/26 | 2 |
| Design radio control override system | Øjan | 100% | 1/16/26 | 1/17/26 | 2 |
| Design electronics box with power system, compute system, and sensor system. | Øjan | 50% | 1/18/26 | 1/19/26 | 2 |
| Assemble electronics into boat | Øjan | 30% | 1/20/26 | 1/25/26 | 6 |
| Documentation | Øjan & Liam | 20% | 1/1/26 | 1/25/26 | 14 |
| **Software interface** | | | | | |
| Research different methods of communication between microcontrollers and Jetson Orin NX | Liam & Øjan | 0% | 2/1/26 | 2/3/26 | 3 |
| Create software for interfacing with the thrusters and servos | Øjan & Liam | 0% | 2/2/26 | 2/5/26 | 3 |
| Setup Ardupilot, connect GPS and CubeOrange. Do some small tests. | Liam | 100% | 1/1/26 | 1/16/26 | |
| Create software for interfacing with the sensors | Liam | 0% | 2/5/26 | 2/8/26 | 4 |
| Create subscriber(s) for the thrusters and servos | Øjan | 0% | 2/8/26 | 2/10/26 | 3 |
| Create a publisher(s) for the sensors | Øjan | 0% | 2/6/26 | 2/9/26 | 4 |
| Documentation | Liam & Øjan | 0% | 2/1/26 | 2/10/26 | 10 |
| **GroundStation (GUI)** | | | | | |
| Implement a method to create points of buoys detected etc. | Liam | 0% | 2/13/26 | 2/23/26 | 8 |
| Create ros2 node to receive and send ros2 topics to the groundstation (May depend on the rest of t | Liam | 0% | 2/11/26 | 2/25/26 | 15 |
| Documentation | Liam | 0% | 2/23/26 | 2/28/26 | 4 |
| **Coordinate Transformations** | | | | | |
| Research how the GPS and IMU should be used to estimate the USV/ASV states | Liam | 0% | 3/2/26 | 3/4/26 | 3 |
| Create service server to convert between Geodetic coordinates to NED and NED to Geodetic. | Liam | 0% | 3/5/26 | 3/7/26 | 3 |
| Create ROS2 topic for subscribing to the interface topics of the sensors. | Liam | 0% | 3/8/26 | 3/9/26 | 2 |
| Create ROS2 node for publishing filtered/processed states | Liam | 0% | 3/10/26 | 3/13/26 | 4 |
| Documentation | Liam | 0% | 3/13/26 | 3/16/26 | 4 |
| **Control System / Dynamic Positioning System** | | | | | |
| Research methods for implementing a DP system | Øjan & Liam | 0% | 2/11/26 | 2/14/26 | 4 |
| Implement simple control system to follow location waypoints | Øjan | 0% | 2/14/26 | 2/18/26 | 5 |
| Experiment and improve control system, on simulator | Øjan | 0% | 2/19/26 | 2/22/26 | 4 |
| Experiment and improve control system, IRL | Øjan | 0% | 2/22/26 | 2/27/26 | 6 |
| [OPTIONAL] System identification, and MPC/LQR | Øjan | 0% | | | |
| Documentation | Øjan | 0% | 2/11/26 | 2/27/26 | 17 |
| **DataLogger*** | | | | | |
| Create logger node | Øjan | 0% | 2/28/26 | 3/4/26 | 5 |
| Subscribe to different topics, and write them to a CSV file | Øjan | 0% | 3/5/26 | 3/8/26 | 4 |
| Documentation | Øjan | 0% | 2/28/26 | 3/8/26 | 9 |
| **Vision System** | | | | | |
| Research frameworks for implementing vision systems(Zed SDK, YOLO & OpenCV) | Liam & Øjan | 0% | 3/17/26 | 3/22/26 | 6 |
| Research methods for implementing object tracking and localization. | Liam & Øjan | 0% | 3/17/26 | 3/22/26 | 6 |
| Find or create dataset for buoys | Øjan | 0% | 3/9/26 | 3/15/26 | 7 |
| Find or create dataset for detection of boats (For collision avoidence). | Øjan | 0% | 3/9/26 | 3/15/26 | 7 |
| Create system for storing and communicating position of detected objects | Liam & Øjan | 0% | 3/23/26 | 3/30/26 | 8 |
| Documentation | Liam & Øjan | 0% | 3/9/26 | 3/30/26 | 22 |
| **Decision System** | | | | | |
| Research on systems to decide on what actions to perform and update gant diagram accordingly | Liam & Øjan | 0% | 3/31/26 | 4/4/26 | 5 |
| Conceptualize procedure for speed mission | Liam & Øjan | 0% | 4/5/26 | 4/15/26 | 11 |
| Conceptualize procedure for obstacle course | Liam & Øjan | 0% | 4/5/26 | 4/15/26 | 11 |
| Conceptualize procedure for docking | Liam & Øjan | 0% | 4/5/26 | 4/15/26 | 11 |
| Conceptualize procedure for avoiding collision with ship | Liam & Øjan | 0% | 4/5/26 | 4/15/26 | 11 |
| Identify commonalities between the task, try to create nodes handling the similar/same features | Liam & Øjan | 0% | 4/5/26 | 4/15/26 | 11 |
| Documentation | Liam & Øjan | 0% | 3/31/26 | 4/15/26 | 16 |
| **Testing** | | | | | |
| Testing and system verification | | 0% | 4/16/26 | 5/4/26 | 19 |
| Documentation | | 0% | 4/16/26 | 5/4/26 | 19 |
| **Project presentations** | | | | | |
| Product pitch video (length: 2 min?) | | | | | |
| Video presentation | | | 5/1/26 | 5/4/26 | 4 |
| Bachelor thesis report | | | 5/5/26 | 5/19/26 | 15 |




*Insert new rows ABOVE this one*

# 6 Documentation

## 6.1 Reports and technical documents

Generally after working with one part of the project, it should be documented by writing further about the method, theory, and result if possible. This is important for ensuring that the other group members can quickly get an idea of how something was developed without going through all of the setup in the software or hardware. The code should include some comments/class description explaining how something was achieved, why and mention potential improvements. The circuits developed should be documented by creating schematics showing the different connections.

# 7 Planned Meetings and Reports

**Meetings with Supervisors** Meetings with the supervisors are scheduled to occur at Tuesday (09:15-10:00) every other week. The meetings will discuss what has been achieved in the sprint period, and the further plans.

**Progress reports** Before each supervision meeting there will be created a progress report. The report states the goals of the periods work, planned activities, achievements and planned activities for the next period. The progress report should not be extremely detailed, but should instead give a high level overview of the achievements, experiences and further plans.

# 8 Planned Deviation-treatment

There are several parts of the project that can go wrong.

**Critical issues with the USV** Since the USV contains a lot of electrical systems, it is important to have an optional plan if the hardware stops working or gets damaged beyond what is recovarable/fixable. If the USV gets severely damaged or is unable to be repaired within a reasonable amount of time the group will focus more on the simulation. In order to be propely prepared for a such scenario it is important to develop a well functioning simulator with the same capabilities as the real vessel.

**Control system/ DP** The azimuth thruster setup can bring a lot of complexity to the project. If controlling and actuating the azimuth thrusters proves to be a more difficult task or if the autopilot for following the coordinates doesn't work well with the azimuth thrusters, there is a plan to lock the thrusters orientation to a given orientation such that the OmniX can be used instead. This allows the group to focus more on the system for actually performing the tasks.

**Critical equipment damage to vision system** If the Zed 2i stereocamera gets damaged/stops functioning then the group will order a new one. Althought expensive, the camera is one of the most important components in the project and must be functioning properly.

# 9 Hardware required

The project relies on a lot of parts which has been ordered previously by Sub-Horizon.

- Zed 2i Camera
- CubeOrangePlus
- Jetson Orin NX 16 GB
- GPS SIMPLERTK3B
- Motor drivers (BlueRobotics Basic ESC)
- T200 thrusters
- High Torque 40 KG Motor Micro Servo
- Radio controller
- Radio receiver
- BlueBoat BaseStation + Antenna
- 2.4gHz Wi-Fi router + Antenna
- Ethernet Switch
- Laptop

# References


[1] Autodrone. *Competition*. URL: `https://www.autodrone.no/competition` (visited on 01/07/2026).

[2] Patryk Cieślak. "Stonefish: An Advanced Open-Source Simulation Tool Designed for Marine Robotics, With a ROS Interface". In: *OCEANS 2019 - Marseille*. June 2019. DOI: `10.1109/OCEANSE.2019.8867434`.